

Applied Functional Analysis Winter 2026

Textbook Applied Analysis by J. Hunter and B. Nachtergaele.

Grading ~ 8 Hrs 30% Midterm + Final 70%.

Motivations for AFA: PDE theory; Proba. theory; Numerical Analysis; M.L.

$x \in X \subset \mathbb{R}^n$ "spatial" variables. $f(x), g(x), u(x)$ "functions".

PDE $\left\{ \begin{array}{l} -\Delta u(x) = f(x) \quad x \in X \\ u(x) = g(x) \quad x \in \partial X \end{array} \right\} \begin{array}{l} \text{spaces of functions?} \\ \Delta: \text{spaces of operators?} \end{array}$

Laplacian operator

boundary condition.

PDE ask: under boundary condition, is there a function u satisfy diff relationship.

Proba. $L^p(X, \mathcal{F}, \mathbb{P})$; SDE $dX = \underbrace{b}_{\text{drift}} dt + \underbrace{\sigma}_{\text{diffusion}} d\underbrace{W}_{\text{brownian motion}} \dots$

Approximations $\partial_x u \approx \frac{1}{h} (u(x + \frac{h}{2}) - u(x - \frac{h}{2}))$. Convergence?

M.L. Complexity of high-dimensional functional spaces

Approximation theories.

SDE and PDE are dual. X_t satisfy SDE $dX_t = b(X_t)dt + \sigma dW_t$. then its density $p(t, x)$ satisfy PDE: $d_t p = -\nabla \cdot (bp) + \frac{1}{2} \sigma^2 \Delta p$

Some terminology / notions

vector does not have size

Topological space $\rightarrow$ no distance
(open sets; continuity)

Measurable Spaces
(σ -algebra of subsets; measure)

Metric Space
(distance)

Vector Space

Normed vector space
(norm)

Convergence

Completeness

Fréchet space
(seminorms)

non zero can also
have size 0
eg: $p(p) = \sup_{s \in [0,1]} |f'(s)|$

Banach Space

(completeness $\Leftrightarrow$ all cauchy converge)

Inner product
(orthogonality)

Hilbert Space

• Topological spaces (Chapter 4 in [HW]).

- Def. Topology: collection $\mathcal{G}$ of open subsets of set X such that:
- (i) $\emptyset$ and X are open; ^{family of} Open sets closed under
 - (ii) arbitrary unions
 - (iii) finite intersections.

→ a rule define open set

$(X, \mathcal{G})$ then a topological space.

A closed $\Leftrightarrow A^c = X \setminus A$ is open.

- Def Neighborhood (nbhd). V is nbhd of x in X if $x \in G \subset V \subset X$ for G open.
can find at least 1 open set G include x

- Def $\mathcal{G}$ is Hausdorff (or separated) if for x, y distinct in X ,
there are nbhds U_x and U_y of x and y s.t. $U_x \cap U_y = \emptyset$.

- Discrete topology: every point open.
 $\mathcal{P}(X) \equiv 2^X$ power set (too big)
- Indiscrete topology: $\emptyset$ and X only open subsets.
(Trivial) (too small)
not Hausdorff if X has ≥ 2 elements.

- Generated topology by subsets $\mathcal{G}_0$: intersection of all topologies $\mathcal{G}$
such that $\mathcal{G}_0 \subset \mathcal{G}$. (Always defined since discrete top. defined).
will make topology be smallest

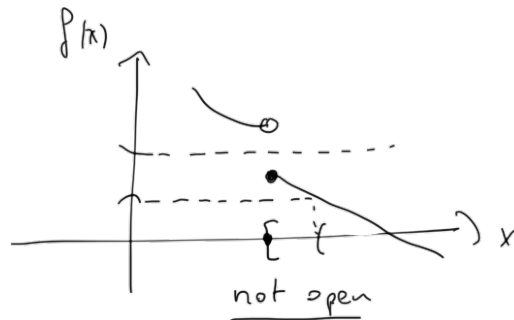
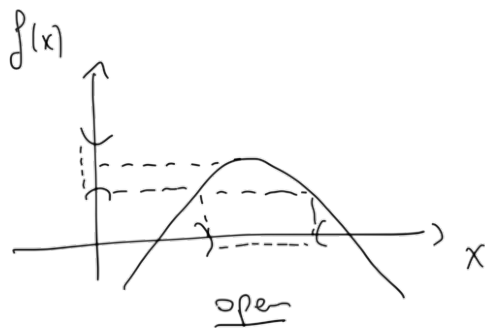
Def Convergence $x_n \rightarrow x$ in X if for all nbhd U_x of x , $x_n \in U_x$ for all n large enough.

Def Continuity $f: (X, \mathcal{C}_X) \rightarrow (Y, \mathcal{C}_Y)$ continuous at x if for all nbhd $W_{f(x)}$ of $f(x)$, there is a nbhd V_x of x s.t. $f(V_x) \subset W_{f(x)}$.

$f: X \rightarrow Y$ continuous if continuous at every $x \in X$.

Theorem $f: (X, \mathcal{C}_X) \rightarrow (Y, \mathcal{C}_Y)$ continuous iff

$f^{-1}(G) \in \mathcal{C}_X$ for all $G \in \mathcal{C}_Y$. $f^{-1}(\text{open})$ is open.



Def $f: X \rightarrow Y$ homeomorphism if bijection and f and f^{-1} are continuous.

X and Y homeomorphic if there is homeomorphism $f: X \rightarrow Y$.

X and Y are then indistinguishable as topological spaces.

Def Compactness $K \subset X$ is compact if every open cover of K admits a finite subcover.
 a seq of open set $\{U_\alpha\}$
 $U_1, \dots, U_n$ s.t. $K \subset U_1 \cup \dots \cup U_n$ s.t. $K \subset \bigcup_\alpha U_\alpha$

On $\mathbb{R}$, topology generated by open intervals $(a, b) = \{x \in \mathbb{R}, a < x < b\}$.

Then $(0, 1)$ not compact. Covered by $\bigcup_{n \geq 1} (\frac{1}{n}, 1 - \frac{1}{n})$. it is a cover by not finite sub cover
 $\rightarrow$ closed + bounded
 $[0, 1]$ is compact (harder to check; follows from Heine-Borel)

Remark It is "easier" to be compact when topology has "less" open sets.

Compactness is one of the main tools providing existence of limits
 (nets have convergent subnets or sequences c.v. subsequences).

compact operators are the ones well approximated by finite-dimensional ones. $\}$ (morally).

. Metric spaces. Include geometry via notion of distance between points in X .

. X non empty set. Def metric (or distance function) on X is $d: X \times X \rightarrow \mathbb{R}$ s.t.

(i): $d(x,y) = d(y,x)$ symmetry; (ii): $d(x,z) \leq d(x,y) + d(y,z)$, x, y, z in X triangle inequality; sub-additivity.

(iii): $d(x,y) \geq 0$ and (iv): $d(x,y) = 0 \iff x = y$.

. Then (X, d) metric space.

. On $\mathbb{R}$, $d(x,y) = |x-y|$.

. on X set of n -letter words in k -alphabet, $d(x,y) = \# \{i \mid x_i \neq y_i\}$ Hamming distance.

. On Cartesian product $d_{X \times Y}((x_1, y_1), (x_2, y_2)) = d_X(x_1, x_2) + d_Y(y_1, y_2)$

$\rightarrow$ distance on $\mathbb{R}^n$ from that on $\mathbb{R}$. *if you have metric, open ball will auto define topology*

. Natural topology $B_\varepsilon(x) = \{y \in X, d(x,y) < \varepsilon\}$ open ball

Then topology on X $\mathcal{C}$ generated by all open balls $B_\varepsilon(x)$; $x \in X$!
 $\varepsilon > 0$!

. closed ball $\overline{B}_\varepsilon(x) = \{y \in X, d(x,y) \leq \varepsilon\}$.

. On $\mathbb{R}$, topology generated by open intervals (a,b) .

• Vector space (Linear space) V over field $\mathbb{F} = \mathbb{R}$ or $\mathbb{F} = \mathbb{C}$ when

$(V, +)$ Abelian group $+ : V \times V \rightarrow V$ $f + g \in V$.
 $\mathbb{F}$ acts on V : $\mathbb{F}$ -module $\cdot : \mathbb{F} \times V \rightarrow V$ $\downarrow f \in V$.

we can add and rescale vectors.

see [HN] for $8 = 2 \times 4$ axioms characterizing group and action structures.

- Examples
- $\mathbb{R}^n$ $x = (x_1, \dots, x_n)$
- $\mathbb{R}^A$: functions from A to $\mathbb{R}$; eg $A = [0, 1]$; $\mathbb{R}$; ...

• $f: (0,1) \rightarrow \mathbb{R}$ $L^2(0,1) : \int_0^1 |f|^2(x) dx < \infty$

$f \in L^2(0,1)$, $g \in L^2(0,1) \Rightarrow f+g \in L^2(0,1)$ | true but needs proof.

like Cauchy Schwarz:

• $C^0([0,1])$: continuous functions on $[0,1]$. $| \langle u, v \rangle |^2 \leq \langle u, u \rangle \cdot \langle v, v \rangle$

- Unit sphere $\{x \in \mathbb{R}^n, |x| = (\sum_{i=1}^n x_i^2)^{1/2} = 1\}$

is not a vector space. It is a metric space

(for natural metric) and hence a topological space (for associated topology).

• Vector spaces are often topological or even metric. Many have "better" structure:

• Norm $\|\cdot\| : V \rightarrow \mathbb{R}$

$$\text{Norm} \left\{ \begin{array}{l} \text{Semi-norm} \end{array} \right\} \left\{ \begin{array}{l} \text{(i): } \|x\| \geq 0 \\ \text{(ii): } \|\lambda x\| = |\lambda| \|x\| \quad \lambda \in \mathbb{R} \text{ (or } \mathbb{F} = \mathbb{R}; \mathbb{C}) \\ \text{(iii): } \|x+y\| \leq \|x\| + \|y\| \quad \text{Triangle inequality} \\ \text{(iv): } \|x\| = 0 \Rightarrow x = 0 \end{array} \right.$$

• Then $(V, \|\cdot\|)$ normed vector space.

• On $(V, \|\cdot\|)$, $d(x, y) = \|x - y\|$ makes (V, d) a metric space.

• For V finite dimensional, i.e. of the form $\mathbb{R}^n$, we will see that all norms are equivalent $\rightarrow \exists C, C > 0 \quad C\|x\|_1 \leq \|x\|_2 \leq C\|x\|_1$,
topology, convergence, are the same

• This is no longer true in infinite dimension, although a natural one makes the space complete.

• notation • $\overline{B} = \{x \in X, \|x\| \leq 1\}$ closed unit ball.

$$B = \{x \in X, \|x\| < 1\} \text{ open}$$

C convex subset when $x, y \in C \Rightarrow tx + (1-t)y \in C; 0 \leq t \leq 1$.

• Example $\mathbb{R}^n$

$$\|x\|_p = \left(\sum_{j=1}^n |x_j|^p \right)^{\frac{1}{p}} \quad 1 \leq p < \infty$$
$$\|x\|_\infty = \sup_{1 \leq i \leq n} |x_i|$$

\begin{array}{l} \vdots \\ \text{norms.} \\ \vdots \end{array}

• Exercise: draw unit balls for $p = 1, 2, \infty$ in $n = 2$.

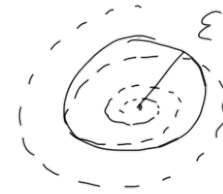
• $\|\cdot\|$ and $\|\cdot\|_1$ are equivalent when $\exists C > 0$ s.t. $C^{-1} \|\cdot\| \leq \|\cdot\|_1 \leq C \|\cdot\|$.

Exercise show all $\|\cdot\|_p$ norms are equivalent on $\mathbb{R}^n$.

$$\exists C_{p,q} > 0 : \|x\|_p \leq C_{p,q} \|x\|_q$$

- Convergence . (Central notion in construction of approximations)

- Def $\{x_n\}_{n \in \mathbb{N}} \in X$ converges to $x \in X$ if $\forall \varepsilon > 0, \exists N = N(\varepsilon) \in \mathbb{N}^*$ s.t.
 $\forall n \geq N, d(x_n, x) \leq \varepsilon$.



- write $x_n \rightarrow x$ as $n \rightarrow \infty$ or $\lim_{n \rightarrow \infty} x_n = x$.

- c.v. in metric sense $\Rightarrow$ c.v. in topological sense.

- Cauchy sequences. Def $(x_n)_{n \geq 1} \in X$ Cauchy if $\forall \varepsilon > 0, \exists N \in \mathbb{N}^*$ s.t.
 $\forall n \geq N$ and $\forall m \geq N, d(x_n, x_m) \leq \varepsilon$.

- no notion of limit there. $x_n \rightarrow x \Rightarrow x_n$ Cauchy $\left\{ \begin{array}{l} d(x_n, x_m) \leq d(x_n, x) + d(x, x_m) \\ \leq \varepsilon/2 \quad \leq \varepsilon/2 \end{array} \right.$

- in $\mathbb{Q}$ $x_n = \frac{p_n}{q_n} \rightarrow \sqrt{2} \notin \mathbb{Q}$. x_n Cauchy not converging in $\mathbb{Q}$.

- This means that $\mathbb{Q}$ is "too small".

- $\mathbb{Q}$ completed $\rightarrow \mathbb{R}$.

too small to include limit pt

• Def (X, d) complete if every Cauchy sequence converges in X .

• A normed vector space that is complete is called a :

Banach Space

• $X \subset \mathbb{R}^n$. Then $C^0(X)$; $C^{k,\alpha}(X)$; $L^p(X)$; $W^{m,p}(X)$ are B-spaces.

• $C^\infty(X) = \bigcap_{k \geq 1} C^k(X)$; $\mathcal{S}(\mathbb{R}^n)$; $\mathcal{S}'(\mathbb{R}^n)$ are Fréchet spaces (not Banach) | some complete metric vector spaces |

• Thm. Every metric space (X, d) has a completion $(\tilde{X}, \tilde{d})$.
 $\forall \tilde{x} \in \tilde{X}, \forall \varepsilon > 0, \exists x \in X$ s.t. $\tilde{d}(x, \tilde{x}) \leq \varepsilon$.
 $X \subset \tilde{X}$; $\tilde{d}(x, y) = d(x, y)$ for $x, y \in X$; X dense in $\tilde{X}$.
 ↳ each new pt can be approach by old pts

• Construction : take $\tilde{x} = (x_n)_n$ a Cauchy sequence.
completion • declare $\tilde{x} \sim \tilde{y}$ when $\lim_{n \rightarrow \infty} d(x_n, y_n) = 0$. Class of equivalence

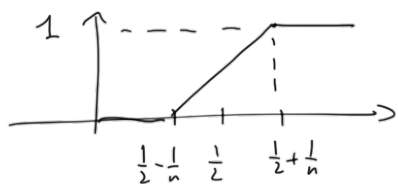
• Define $\tilde{X}$ as set of all classes of equivalence
 and $\tilde{d}(\tilde{x}, \tilde{y}) = \lim_{n \rightarrow \infty} d(x_n, y_n)$

• Prove $(\tilde{X}, \tilde{d})$ complete and an extension of (X, d) .

• Completion unique up to isomorphism.

• Example $\mathcal{C}([0,1])$ continuous functions. $\|u\|_2 = \left(\int_0^1 |u|^2(x) dx \right)^{1/2}$.

$(\mathcal{C}, \|\cdot\|_2)$ is normed vector space. But not complete:



Cauchy sequence for $\|\cdot\|_2$
but limit not in $\mathcal{C}$

• Completion of $(\mathcal{C}, \|\cdot\|_2)$ is $(L^2(0,1); \|\cdot\|_2)$ fng if f=g a.e.

space of square-integrable "functions". In fact classes of

equivalence of functions defined up to set of measure 0

(since $L^2(0,1) = L^2((0,1), \mathbb{F}, d)$ measure space).

• Completion of $(\mathcal{C}, \|\cdot\|_p)$ is $(L^p, \|\cdot\|_p)$ for $1 \leq p < \infty$

• $\mathcal{C}([0,1], \|\cdot\|_\infty)$ is Banach with uniform norm.

• Continuity in metric spaces (X, d_X) ; (Y, d_Y) .

• Def: $f: X \rightarrow Y$ continuous at $x_0 \in X$ if $\forall \varepsilon > 0, \exists \delta = \delta(\varepsilon)$ s.t. $d_X(x, x_0) < \delta \Rightarrow d_Y(f(x), f(x_0)) < \varepsilon$.

• Def: $f: X \rightarrow Y$ continuous if continuous at each $x \in X$.

• Above $\delta = \delta(\varepsilon; x_0)$ depends on x_0 . When $\delta(\varepsilon)$ independent of x_0 , f is uniformly cont.

• Sequential continuity f seq. cont. at x if $(x_n \rightarrow x) \Rightarrow (f(x_n) \rightarrow f(x))$.

• Proposition $f: X \rightarrow Y$ continuous iff sequentially continuous. (Exercise)

• Proposition $F \subset X$ closed iff $\left\{ (x_n \rightarrow x) \Rightarrow x \in F \mid \substack{x_n \in F; x \in X} \right\}$.

converged seq from F also converge to x in F

• Closure $\bar{A}$ of A in X = smallest closed set including A

$$\bar{A} = \{ x \in X, \exists x_n \in A, x_n \rightarrow x \}$$

• Density $A \subset X$ is dense when $\bar{A} = X$; $\bar{\mathbb{Q}} = \mathbb{R}$.

(X, d) dense in its completion $(\tilde{X}, \tilde{d})$ (for metric $\tilde{d}$!).

• (X, d) separable if it has countable dense subset. Example $\mathbb{R} = \bar{\mathbb{Q}}$; $\mathbb{Q} \sim \mathbb{N}$.

- Compactness . Essential notion of existence.
- Def $K \subset X$ is sequentially compact if every sequence in K admits a converging subsequence in K .
- Subsequence : $\varphi: \mathbb{N}^* \rightarrow \mathbb{N}^*$ such that $\varphi(n+1) \geq \varphi(n) + 1$.
↙ index function
- $X_n = (-1)^n$ does not converge. X_{2n} and X_{2n+1} are converging subsequences.
- Theorem $K \subset X$ metric is compact iff it is sequentially compact.
if metric K compact iff K seq compact
- A is called precompact when $\bar{A}$ is compact.
- $K \subset X = \mathbb{R}$.
 . K unbounded $\Rightarrow$ not compact
 . $K = [0, 1]$ not compact . $K = [0, 1]$ compact (needs a proof).
- Prop. K compact $\Rightarrow$ K bounded and closed (Exercise)

• Theorem. Heine-Borel. Subsets in $\mathbb{R}^n$ are compact if and only if they are closed and bounded.

• Theorem Bolzano-Weierstrass: every bounded sequence in $\mathbb{R}^n$ admits a convergent subsequence.

Exercise: BW $\Rightarrow$ H.B. Derivation of B.W. $\mathbb{R}^2$. x_n bounded in $[-1,1]^2 = S_0$



At each step $k \geq 1$, divide by 2 in each direction and keep 1/4 of domain with infinite number of points x_n , called S_k .

"Keeping" means introducing subsequence φ_k .

Then $x_{\varphi_k(0) \dots \varphi_k(n)} \in S_k \subset S_0$ with S_k of size $\frac{1}{2^{k-1}}$.

• Diagonal extraction $\xi_k = x_{\varphi_k(0) \dots \varphi_k(k)}$. Check ξ_k subsequence of x_n .

• ξ_k is Cauchy: $\left| \xi_{k+m} - \xi_k \right| \leq \frac{C}{2^{k-1}} \rightarrow 0$ as $k \rightarrow \infty$ for all $m \geq 0$

$\bigwedge_{S_k} \quad \bigwedge_{S_k}$

• $\mathbb{R}^n$ complete $\Rightarrow \xi_k \rightarrow \xi \in S_0$ $\square$

- H.B. does not apply in infinite dimensions. In $\ell^2 = \{x = (x_n)_{n \in \mathbb{N}}, (\sum_{n=1}^{\infty} x_n^2)^{1/2} < \infty\}$, construct "basis" $e^k \in \ell^2$ such that $e_j^k = \delta_{jk} = \begin{cases} 1 & j=k \\ 0 & j \neq k \end{cases}$. $e^k \in \overline{B}$ closed unit ball.
- choose $x_k = e^k$ for $k \geq 1$. $x_k \in \overline{B}$ bounded. $\|e^k - e^l\|_2^2 = 2$ when $k \neq l$.
- For any subsequence, $\|x_{\varphi(k)} - x_{\varphi(l)}\|_2^2 = 2$ fixed not Cauchy so not convergent.

• Abstract result for arbitrary compact subsets of metric spaces:

• Def $\{G_\alpha, \alpha \in A\}$ cover of A if $A \subset \bigcup_{\alpha \in A} G_\alpha$. Ball

• Def $\{x_\alpha, \alpha \in A\}$ is ε -net of A if $A \subset \bigcup_{\alpha \in A} B_\varepsilon(x_\alpha)$.

• Def $A \subset X$ is totally bounded if it has a FINITE ε -net for all $\varepsilon > 0$.

Theorem $A \subset X$ is sequentially compact iff it is complete and totally bounded.